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Adrenal

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Bilateral Adrenal Infarcts: Initial Presentation in Concurrent Myelodysplastic Syndrome and Anti-Phospholipid Syndrome

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Introduction: Bilateral adrenal hemorrhage (BAH) has been noted in several conditions. Bilateral adrenal infarcts (BAI) are rarer but have been seen in anti-phospholipid syndrome (APS). Adrenal insults are not commonly associated with myelodysplastic syndrome (MDS). We present a patient who developed BAI as initial presentation of APS in the setting of MDS and discuss management considerations.

Case: A 58-year-old female with MDS diagnosed one month prior presented with abdominal pain, present for one year but acutely worse in the past week. CT A/P showed nodularity of bilateral adrenal glands. She was treated conservatively but continued to experience nausea, vomiting, and abdominal pain – repeat CT A/P one month later was suspicious for bilateral peri-adrenal hemorrhages. MRI with adrenal protocol demonstrated bilateral non-hemorrhagic adrenal infarcts. VS were negative for hypothermia or hypotension. Labs showed Na 136 (135-145 mmol/L), K 3.5 (3.4-5.1 mmol/L), BG 108 (mg/dL), DHEA-S 9.3 (8.0-391 mcg/dL), renin 6.7 (0.5-4.0 ng/mL/hr), aldosterone 6.2 (<31.0 ng/dL), 60-minute cortisol after ACTH stimulation test 34.3 (5.2-22.5 mcg/dL), plasma metanephrines <0.10 (0.00-0.49 nmol/L), plasma normetanephrines 0.34 (0.00-0.89 nmol/L), Hgb 11.8 (12.0-15.5 g/dL), and plts 129K (140-450 K/mcL). No history of anticoagulant use, trauma, or VTE. COVID-19 PCR was negative on two separate tests. Hypercoagulability testing noted positive lupus anticoagulant. Anticoagulation was deferred due to underlying bone marrow abnormalities and thrombocytopenia; aspirin was recommended instead by hematology. Outpatient testing revealed 12pm cortisol of 12.6 (5.2-22.5 mcg/dL), ACTH 108.3 (7.2-63.0 pg/mL), and DHEA-S 6.2 (8.0-391 mcg/dL). Patient was empirically started on hydrocortisone for glucocorticoid replacement therapy.

Conclusion: BAI is not typically associated with MDS but can occur in the setting of a concurrent hypercoagulable process such as APS. The adrenal glands are the most commonly involved glands in APS. Patients often present in adrenal crisis but can present in stable condition as in this case. The adrenal glands are particularly vulnerable due to the "vascular dam" near the zona reticularis. Imaging and laboratory investigation can clarify the nature of the adrenal insult and guide management. Treatment strategy may include anticoagulation to prevent further VTE events, but bleeding risk must be considered. Both glucocorticoid and mineralocorticoid replacement should be considered until residual adrenal reserve can be fully elucidated, which may take several months from initial diagnosis.

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